

EFF-Dock: Fragment-Based SE(3)-Equivariant Flow Matching for Protein–Ligand Docking

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Abstract

Protein–ligand docking requires accurate candidate generation, physically plausible coordinates and reliable pose selection. We present EFF-Dock, a fixed-receptor, supplied-pocket method that learns a flow over the translations and rotations of rigid ligand fragments. An equivariant protein–ligand graph network predicts atom-wise vectors that are projected onto fragment motion, a separate network ranks candidates, and a differentiable interaction energy refines their geometry. Across five reference-pocket redocking cohorts, refined 100-pose banks contained a candidate with RMSD below 2 Å for 87.9–96.5% of complexes. Refinement followed by rescoring increased selected-pose RMSD–PoseBusters success by 15.7–23.2 percentage points under the primary chirality-filtered selector (21.7–26.5 without filtering), giving success rates of 53.2–80.0%. In 12.4–34.4% of complexes, the eligible bank contained an RMSD-successful candidate but the selected pose was not RMSD-successful. These analyses distinguish candidate availability, coordinate validity and ranking loss, while pocket-center perturbations expose generation-input sensitivity with fixed downstream spatial context.

The three-dimensional pose of a small molecule determines its protein contacts and shapes subsequent structure–activity analysis. Experimental complexes provide the best reference, but they are unavailable for many targets and ligand series. Protein–ligand docking fills this gap by predicting ligand placement and conformation for a prepared receptor. Here we study fixed-receptor docking with a supplied binding site; pocket discovery, binding classification, and affinity prediction are outside the scope of the model.

Docking remains difficult because global placement and internal conformation are coupled on a rugged, multimodal landscape. A finite search must resolve translation, orientation, rotatable bonds, covalent geometry, and local protein–ligand complementarity. Protonation and tautomeric state, cofactors, metals, solvent, and receptor reorganization create additional ambiguity that is not removed by increasing the number of sampled poses. Classical docking programs combine conformational search with approximate scoring [1, 2]. GNINA augments this workflow with learned three-dimensional convolutional scoring [3]. Their output depends strongly on receptor preparation, search-region definition, flexibility settings, and post-docking refinement.

Learned geometric models have broadened both the search mechanism and the task definition. DiffDock samples ligand translation, rotation, and torsion by a diffusion process [4]; Uni-Mol Docking V2 predicts poses within a specified binding pocket [5]; and DiffBindFR jointly treats ligand motion and selected receptor-side-chain flexibility [6]. DynamicBind generates ligand-specific receptor conformations [7], FlowDock applies conditional geometric flow matching to flexible complex generation [8], and AlphaFold 3 addresses a broader biomolecular cofolding problem [9]. These methods solve different tasks and receive different receptor and pocket information. Comparisons are restricted to supplied-pocket protocols and retain each method’s native receptor handling; cofolding methods are discussed as broader tasks.

A docking pipeline has three main components: pose generation, candidate ranking, and coordinate validation. Oracle minimum RMSD over a finite candidate set measures sampling coverage, not top-

1 docking. A low ligand RMSD also does not guarantee valid stereochemistry, bond geometry, or protein–ligand contacts [10]. We report coverage, selected top-1 accuracy, PoseBusters validity, and RMSD–PoseBusters success as separate endpoints. A confidence network is used only for ranking within a candidate bank, consistent with recent work that separates pose quality from binding-oriented predictions [11].

Ligand flexibility can be parameterized at several geometric scales. Treating the ligand as one rigid body preserves internal geometry but cannot represent torsional rearrangement. Direct all-atom generation is flexible but requires the model to maintain covalent structure while solving global placement. Torsion-tree methods preserve local bond geometry, although each rotation moves a convention-dependent downstream subgraph. Classical incremental docking instead anchors and grows rigid ligand pieces [12], while chemistry-oriented schemes such as Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) define fragments for a different purpose: retrosynthetically meaningful molecular decomposition [13]. SigmaDock applies diffusion to rigid-fragment ligand poses [14]. Our decomposition cuts a restricted set of rotatable bonds to define rigid components that move simultaneously. It lies between a single rigid ligand and unconstrained atom coordinates, while soft geometric terms maintain the cut interfaces.

Flow matching learns a continuous vector field that transports a tractable prior toward a data distribution without integrating the generative ordinary differential equation (ODE) during training [15]. Its extension to non-Euclidean state spaces provides a natural language for translation–rotation products [16]. Matcha applies multistage Riemannian flow matching to molecular docking [17]. Equivariant tensor-product networks allow scalar, vector, and higher-order features to transform consistently under a joint rigid motion [18]. Related flexible-docking work has used Newton–Euler mechanics to map interaction energy to geometric motion [19]. In EFF-Dock, a shared protein–ligand network instead predicts atom-level equivariant vectors and aggregates them for each ligand fragment. An independently parameterized interaction energy provides pose diagnostics, optional inference guidance, and post-sampling refinement.

EFF-Dock represents the ligand on a product of fragment rigid-body pose spaces. A heterogeneous graph connects ligand atoms and fragments with protein atoms and residue nodes through covalent, topological, and spatial relations. A shared SE(3)-equivariant trunk and fragment-wise Newton–Euler readout define the conditional flow, while a separate confidence model ranks the sampled poses.

Here we evaluate candidate availability, physical validity and pose selection separately across five reference-pocket redocking cohorts. The experiments assess the resulting pipeline rather than isolating the causal contribution of each architectural component.

1 Results

1.1 Fragment-space docking pipeline

EFF-Dock represents a ligand as rigid components obtained by cutting a restricted set of rotatable bonds. A heterogeneous protein–ligand graph conditions a flow over the fragments’ translations and rotations (Fig. 1). An equivariant network predicts one vector per ligand atom; a least-squares rigid-motion projection converts these vectors to fragment velocities (Fig. 2). A separate network predicts a within-bank pose-ranking score. Generated candidates are refined by bounded descent on a differentiable geometric interaction energy and then ranked again. The energy supplies coordinate gradients independently of the learned vector field; it is neither an affinity predictor nor a free-energy model.

The evaluated pipeline uses a fixed holo receptor, a reference-derived pocket, 100 candidates and ten integration steps. Both networks are held fixed across three inference seeds. These repeats quantify variation in the executed inference protocol and do not represent independent model training. Evaluation revises local-to-world feature injection and coordinate-to-frame recovery relative to training without retraining either network. Compared with the archived pipeline, selected RMSD and RMSD–PoseBusters rates changed by -0.3 to $+0.2$ percentage points on Astex, PoseBusters and PhiBench-derived; these comparisons do not isolate a component effect (Supplementary Information, Sec. S1.2). The complete input and coordinate conventions are specified in Online Methods.

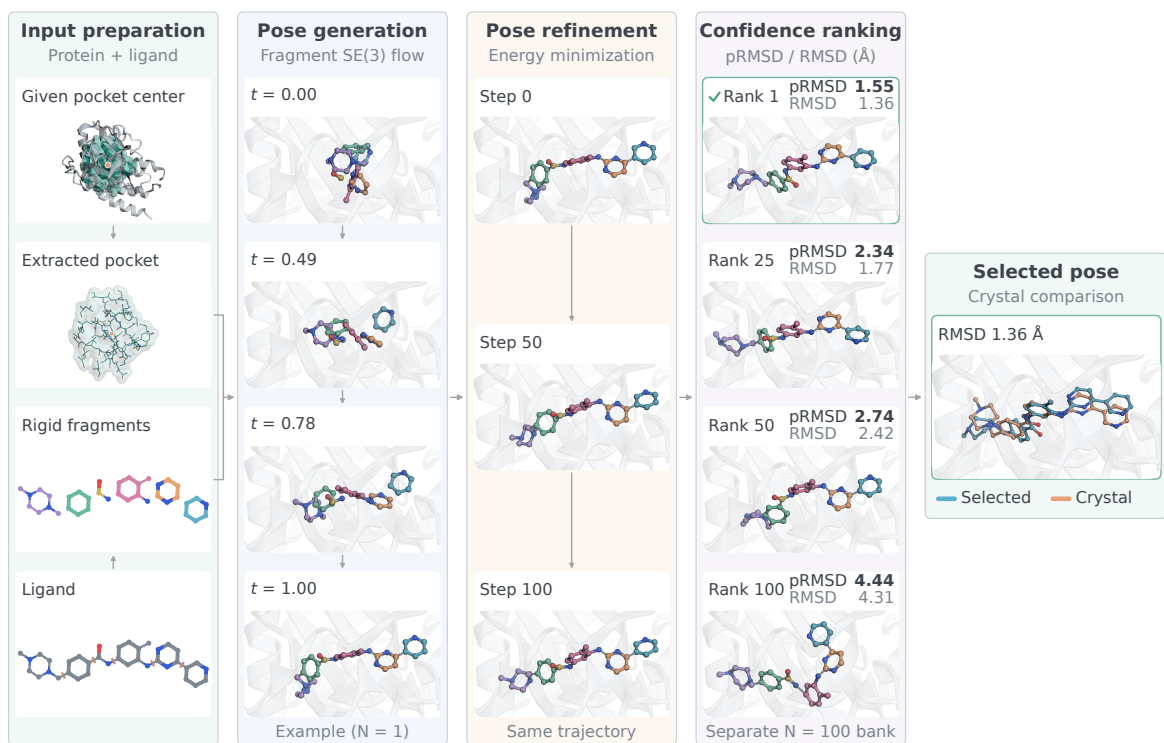
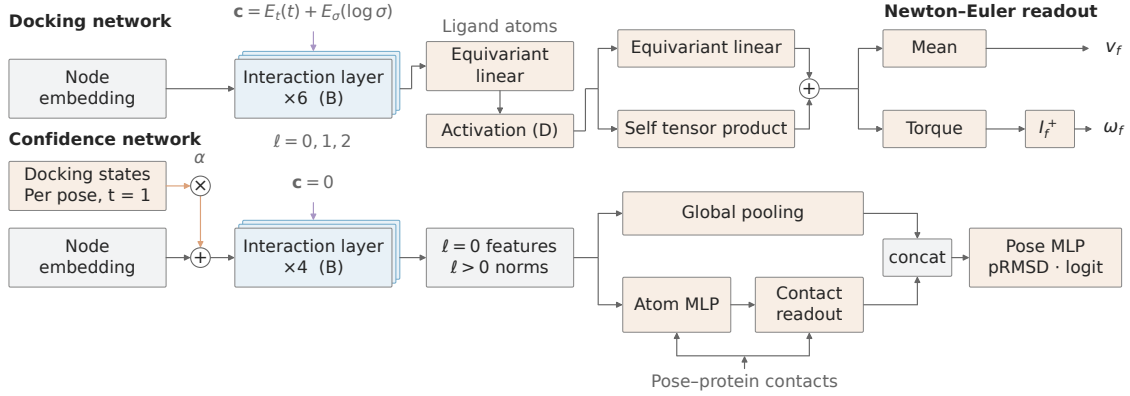
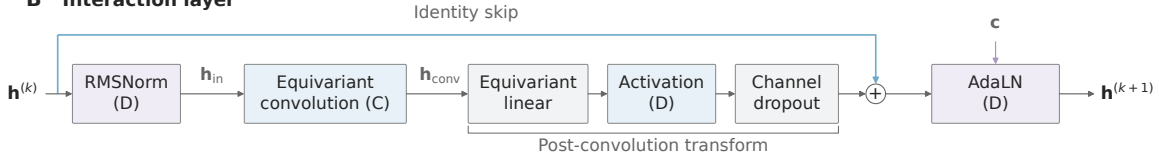


Figure 1: **Overview of the EFF-Dock docking pipeline.** A prepared receptor, one ligand chemical state and conformer, and a supplied pocket center define the input. The ligand is decomposed into rigid fragments whose SE(3) states are transported by the learned flow. The illustrated candidate is subsequently optimized by bounded rigid-fragment energy refinement. A separate 100-pose benchmark bank prepared from SMILES is scored by the confidence model, and the selected pose is compared with the crystal pose only for evaluation. The checkmark identifies the selected candidate; crosses mark unselected examples. Candidate ranks use predicted RMSD (pRMSD); the displayed RMSD values are retrospective errors and do not enter selection. The illustrated trajectory uses a supplied starting conformer and is a prespecified one-pose example, not an aggregate benchmark result.

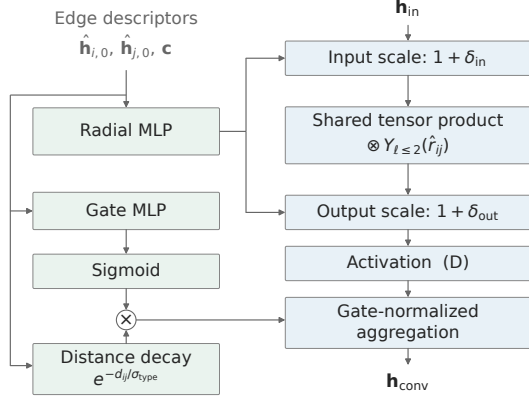
A Model architecture



B Interaction layer



C Equivariant convolution



D Normalization and activation

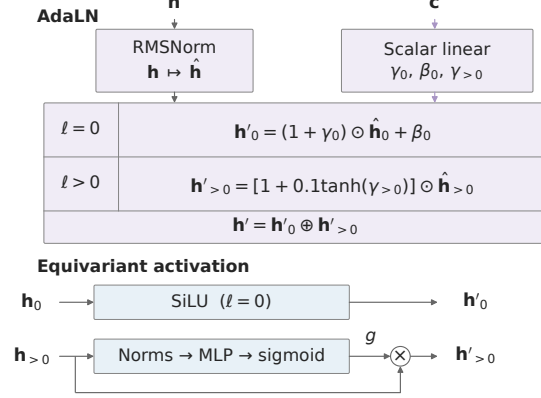


Figure 2: Docking and confidence architectures. (A) The docking network applies six conditioned equivariant interaction layers and converts ligand-atom features to fragment translation and angular velocities with the Newton–Euler readout. For each candidate, terminal docking features are recomputed at $t = 1$ with its source prior scale. The confidence network combines these with a new embedding, applies four interaction layers with zero conditioning vector $\mathbf{c} = 0$, invariantizes the resulting features, and predicts pose-level RMSD and success scores from global and contact-aware summaries. (B) Each interaction layer contains normalization, equivariant convolution, post-convolution transformation, a residual connection, and adaptive layer normalization. (C) The equivariant convolution uses radial and gating networks, shared tensor products through $\ell \leq 2$, distance decay, and gate-normalized aggregation with invariant nonscalar rescaling. (D) Scalar and nonscalar channels use type-appropriate adaptive normalization and equivariant activation. Detailed tensor definitions are provided in Supplementary Information, Sec. S3.

1.2 Supplied-pocket comparison

Fig. 3 compares selected-pose performance on Astex Diverse and PoseBusters v2. The local comparison includes SigmaDock [14], DiffDock-Pocket [20], RLDiff RL++ [21], DiffBindFR [6], and force-optimized SurfDock [22]. On Astex, EFF-Dock achieved 83.1% RMSD success and 80.0% same-pose RMSD–PoseBusters success. SigmaDock, SurfDock and RLDiff RL++ had higher point estimates on both endpoints. On PoseBusters v2, EFF-Dock reached 81.5% RMSD success and 79.1% RMSD–PoseBusters success. Relative to SigmaDock, the paired difference in RMSD–PoseBusters success was +2.71 percentage points (95% bootstrap interval, -2.27 to $+7.47$), compared with -9.02 points (-16.08 to -1.96) on Astex. The PoseBusters interval does not establish superiority, equivalence or non-inferiority; intervals are unadjusted for multiple comparisons (Supplementary Fig. S14). Requiring PoseBusters validity reduced RMSD-only success by 3.1 percentage points on Astex and 2.4 points on PoseBusters. Most selected RMSD-successful EFF-Dock poses therefore passed the full validity endpoint. The learned methods use their native sampling and ranking procedures, while GOLD and Vina are literature values [10]; the comparison evaluates protocol-conditioned pipelines with differing receptor and ligand preparation, pocket construction, candidate budgets and native selectors, rather than matched-compute implementations. Saved-pose reevaluation under benchmark-specific endpoints gave 71.4% top-1 success on all 558 FoldBench interfaces, requiring BiSyRMSD < 2 Å and LDDT-PLI > 0.8 on the same refined pose. On the official 802-case OpenBind follow-on subset, top-1 BiSyRMSD ≤ 2 Å and PoseBusters success was 56.4%, falling to 48.3% with the additional LDDT-PLI ≥ 0.8 requirement (Supplementary Fig. S20; [Supplementary Information, Sec. S7.3](#)). PhiBench-derived results retain the local cohort and fixed-receptor RMSD definition. Published results accompany these analyses, but differences in tasks, cohorts, preparation, training exposure and candidate budgets preclude direct paired comparisons.

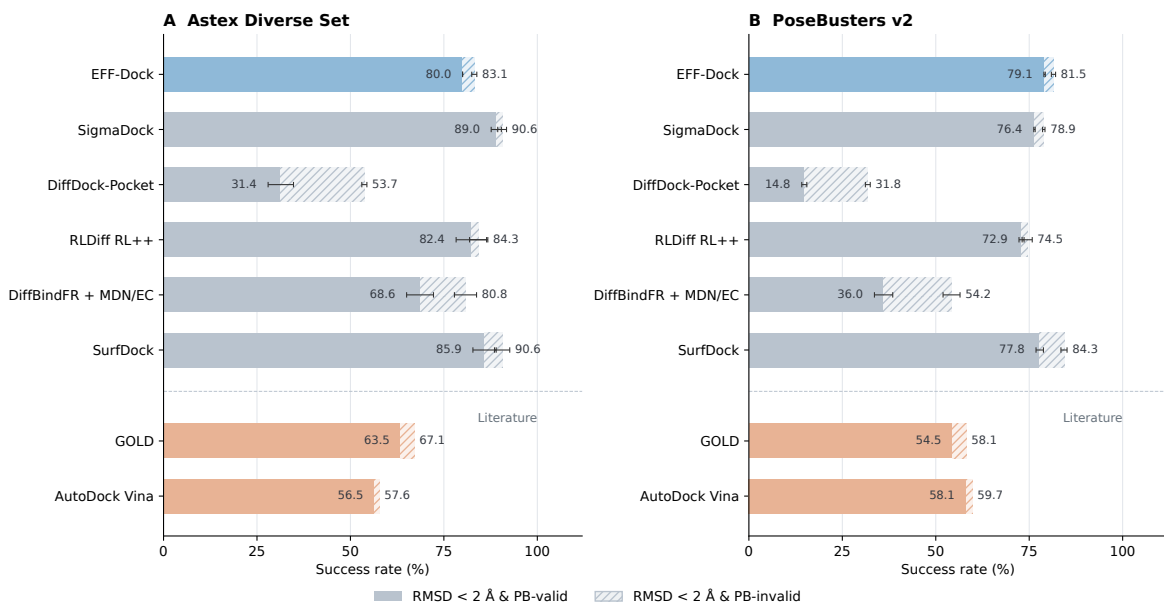


Figure 3: **External docking performance on supplied-pocket benchmarks.** (A) Astex Diverse Set (85 complexes) and (B) PoseBusters v2 (308 complexes). Solid bars show RMSD < 2 Å and PoseBusters success; hatched segments complete RMSD-only success. Local results are means with sample SD across three repeats. EFF-Dock uses 100 poses, 10 sampling steps, refinement, and chirality-filtered confidence selection. GOLD and AutoDock Vina are literature values; the remaining methods use their native sampling and ranking protocols. DiffDock-Pocket uses a holo-aligned predicted receptor; DiffBindFR samples pocket side-chain torsions on a fixed backbone and uses cognate-ligand coordinates in its upstream scoring graph; its selected ligand poses are evaluated against the common frozen holo receptor. Budgets, receptor preparation, selectors and evaluation versions differ, so these are protocol-conditioned comparisons.

1.3 Refinement and chirality-aware selection

Fig. 4 compares coordinate-refinement and chirality-aware selection conditions. Without the chirality filter, refinement increased RMSD–PoseBusters success from 54.5% to 77.6% on Astex, 51.3% to 77.8% on PoseBusters, 37.2% to 58.9% on PhiBench-derived, 49.0% to 72.9% on FoldBench, and 26.9% to 52.9% on OpenBind. Without chirality filtering, these are gains of 21.7–26.5 percentage points across the five cohorts; the corresponding gains under the primary chirality-filtered selector were 15.7–23.2 points. Mean RMSD-only selected-pose success was 1.0–3.7 points higher after refinement. These pipeline-level differences combine coordinate changes with changes in the selected candidate after fresh confidence scoring; they do not measure the repair of a fixed pose.

After refinement, chirality filtering increased RMSD–PoseBusters success by 2.4 points on Astex, 1.3 on PoseBusters, 1.6 on PhiBench-derived, 0.8 on FoldBench, and 0.3 on OpenBind (Supplementary Fig. S9). The resulting primary success rates were 80.0%, 79.1%, 60.5%, 73.8%, and 53.2%, respectively. The refinement-and-rescoring stage accounts for most of the observed gain; chirality filtering adds a smaller improvement. Across all candidates, input-tetrahedral compatibility increased from 58.5–74.4% in raw banks to 88.2–96.8% after refinement. Rigid fragments preserve their own geometry but do not automatically preserve stereochemistry across cut interfaces.

The complexity analysis in Supplementary Fig. S1 localizes the remaining error. Selected top-1 and RMSD–PoseBusters success generally deteriorated for larger ligands, more rotatable bonds, and more fragments, most clearly in PoseBusters, PhiBench-derived, and FoldBench. Oracle success was substantially less sensitive than selected top-1, so the top-1–oracle gap tended to widen in harder strata. Molecular complexity affects ranking and selected-pose validity as well as near-native pose generation. The smallest Astex and high-complexity bins are too sparse for a clear trend.

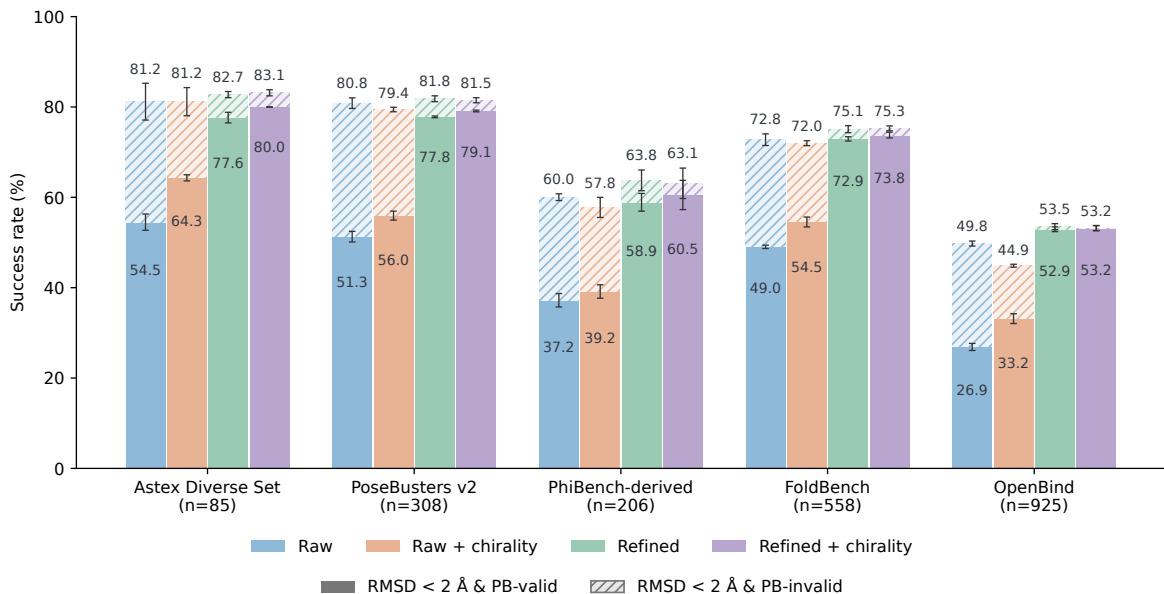


Figure 4: **Effects of refinement and chirality-aware selection.** EFF-Dock results are shown for Astex, PoseBusters v2, PhiBench-derived, FoldBench, and OpenBind using the frozen released weights and unguided 100-pose, 10-step sampling. Raw and refined banks are ranked with or without the chirality filter. Solid bars show RMSD and PoseBusters success; hatched segments complete RMSD-only success. Bars and error bars are the mean and sample SD across three repeats on fixed cohorts.

Supplementary Fig. S16 resolves the aggregate validity gain into individual PoseBusters checks. Refinement reduced selected-pose failures associated with bond geometry, internal clashes, receptor clashes, and internal energy, while stereochemical and ring-planarity failures were less uniformly affected. The matched-candidate panels separate same-index fail-to-pass and pass-to-fail transitions from changes caused by selecting a different pose after refinement. Because matched pairs exist only for candidates with saved official evaluations at both stages, these transitions diagnose the evaluated subset rather than the complete candidate banks.

1.4 Confidence ranking and selection headroom

Fig. 5 evaluates the ranker within each generated bank. For refined banks, mean within-complex Spearman correlation between predicted and observed RMSD ranged from 0.51 to 0.70; near-native classification AUROC ranged from 0.88 to 0.94 and average precision from 0.61 to 0.82. The median selection regret and conditional top-1 success in Fig. 5C,D quantify how an available RMSD-successful candidate can fail to be selected. Reliability curves and performance as a function of near-native candidate density are reported in Supplementary Fig. S12.

Confidence ranking nevertheless enriches near-native poses at early ranks relative to saved generation order (Supplementary Fig. S2). For the refined unfiltered banks, top-1 RMSD success was 82.7%, 81.8%, 63.8%, 75.1%, and 53.5% on Astex, PoseBusters, PhiBench-derived, FoldBench, and OpenBind, respectively. Expanding from top-1 to top-5 raised coverage to 92.9%, 89.3%, 78.2%, 85.4%, and 73.8%—gains of 7.5–20.3 points from only four additional ranked candidates. The confidence score enriches useful poses at early ranks, although rank one remains imperfect.

The corresponding oracle-100 values were 96.5%, 94.3%, 88.5%, 91.6%, and 87.9% (Supplementary Fig. S10). The remaining top-5-to-oracle gaps were 3.5, 5.0, 10.4, 6.2, and 14.0 points, respectively. OpenBind was the most selection-limited cohort: only 6.3% of its individual refined candidates were near-native, yet 87.9% of complexes contained at least one such pose. The corresponding candidate densities were 34.5% for Astex, 31.7% for PoseBusters, 22.9% for PhiBench-derived, and 26.0% for FoldBench.

The generation-prefix experiment in Supplementary Fig. S3 reaches the same conclusion from a deployable selection perspective. Oracle coverage cannot decrease as additional generated poses become available, whereas confidence-selected top-1 can plateau or decline when a newly added pose is incorrectly promoted. Chirality filtering improves some prefixes but does not remove this non-monotonicity. Larger candidate banks improve coverage, but the gain reaches top-1 only when the ranker remains stable as new poses are added.

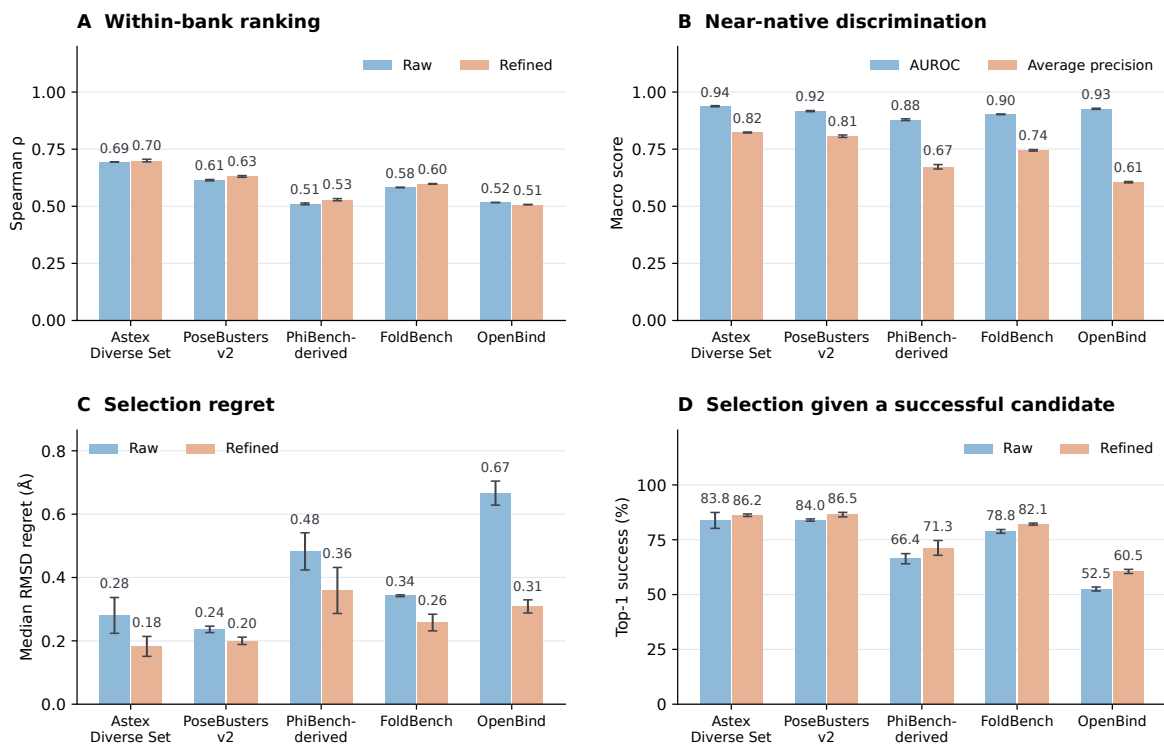


Figure 5: **Confidence ranking and remaining selection loss.** (A) Within-complex Spearman correlation between predicted and observed RMSD for raw and refined banks. (B) Refined-bank AUROC and average precision for strict RMSD < 2 Å labels; single-class banks are excluded. (C) Median selected-minus-full-oracle RMSD regret. (D) Selected top-1 success conditional on the full bank containing a near-native candidate. Panels C and D use the primary chirality-filtered selector, so their loss can include eligibility exclusions. Bars show three-repeat means and sample SD.

1.5 Eligibility-aware selection bottlenecks

Supplementary Fig. S18 partitions every primary outcome into absence of an RMSD-successful candidate in the refined bank, exclusion of all near-native candidates by the effective chirality mask, ranking failure within the eligible set, PoseBusters failure of an RMSD-successful selection, or RMSD–PoseBusters success. The fraction without an RMSD-successful refined candidate ranged from 3.5% on Astex to 12.1% on OpenBind, whereas chirality exclusion accounted for at most 0.3%. Ranking failure was larger, ranging from 12.4% on PoseBusters to 34.4% on OpenBind. Selected RMSD-successful poses failing PoseBusters accounted for 0.0–3.1% of all complexes. Within the refined banks, RMSD-based ranking failure exceeds chirality exclusion. Because PoseBusters was not evaluated for every candidate, this gap does not quantify attainable improvement in RMSD–PoseBusters success. The regret curves in Supplementary Fig. S18B measure loss relative to the best RMSD in the same effective candidate set available to the deployed selector.

Training-set relatedness is examined in Supplementary Figs. S6–S8. Sequence and ligand overlap differed substantially among cohorts, but neither exact ligand matches nor sequence-identity strata showed a consistent association with docking success. These analyses qualify claims of prospective generalization without identifying a single overlap measure that explains the benchmark results.

1.6 Sensitivity to generation inputs

We varied the generation crop, supplied-center error and translation-prior scale on all Astex and PoseBusters complexes while holding the networks, downstream spatial context and primary selector fixed (Fig. 6). With a 10 Å crop and prior scale 2 Å, RMSD–PoseBusters success was 80.0% on Astex and 79.1% on PoseBusters without center error. Gaussian center error with standard deviation 1 Å per axis gave 75.3% and 76.3%, respectively; at 2 Å, success was 71.0% and 64.4%. Refined-bank RMSD-only oracle coverage at 2 Å error was 86.3% and 82.8%.

At a 6 Å crop with 2 Å center error, RMSD–PoseBusters success was 66.3% on Astex and 55.1% on PoseBusters. A 14 Å crop under the same error gave 71.4% and 65.5%. At the unperturbed 10 Å crop, prior scales of 1/2/4 Å gave RMSD–PoseBusters success of 83.9/80.0/74.5% on Astex and 79.8/79.1/75.1% on PoseBusters. These descriptive comparisons do not select a new operating point.

Refinement and confidence scoring retained the original unperturbed center and a 10 Å crop in all conditions. The crop and center-jitter grids perturb sampling inputs with fixed downstream spatial context. Prior-scale comparisons include the scale’s influence on both sampling and the recomputed docking features used for ranking. Guidance and candidate-allocation controls at equal learned pose-step counts are reported separately (Supplementary Fig. S19).

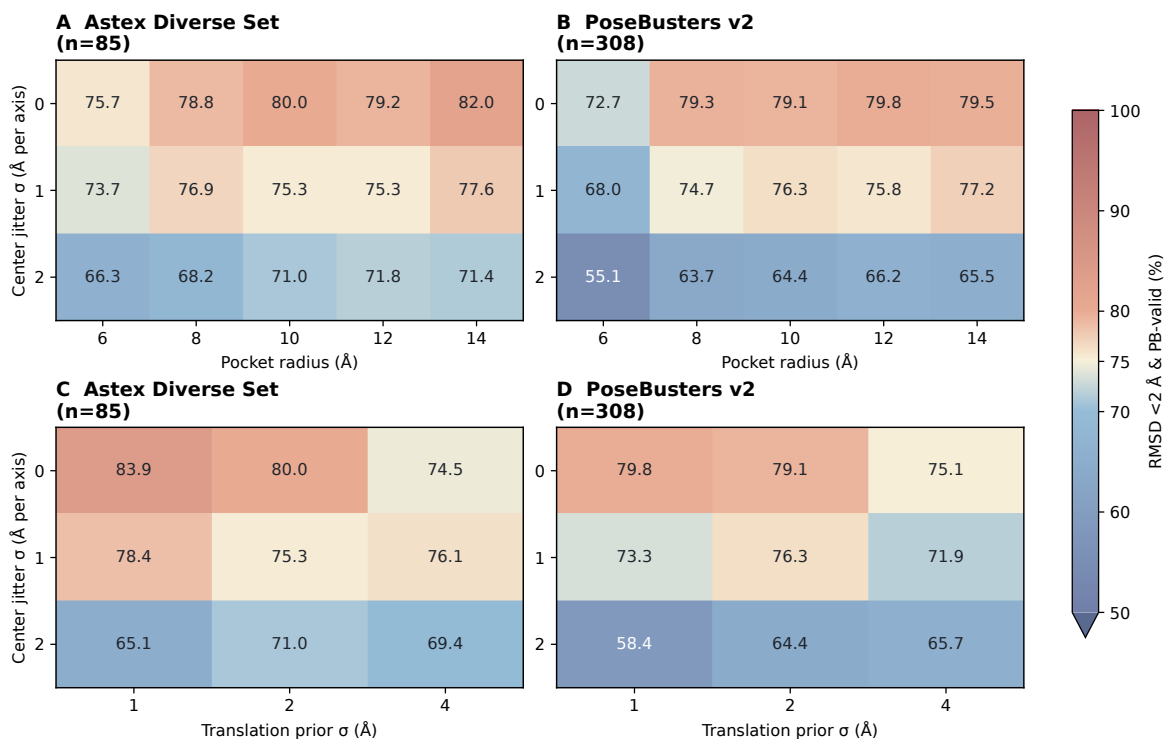


Figure 6: **Sensitivity to supplied-pocket radius, center error and translation-prior scale.** (A,B) Joint selected-pose RMSD < 2 Å and PoseBusters success across generation crop radii at prior scale 2 Å, for Astex Diverse Set (85 complexes) and PoseBusters v2 (308 complexes). (C,D) Prior-scale dependence at a 10 Å generation crop on the same cohorts. Rows vary Gaussian center error, with standard deviation reported per Cartesian axis. Cells show three-inference-seed means. All conditions use unguided 100-pose, ten-step generation, energy refinement and chirality-filtered confidence ranking. Refinement and ranking retain the original center and 10 Å crop; terminal docking features used for confidence retain the generation prior scale. The grid isolates the crop and prior axes rather than crossing every pair. Downstream spatial context is fixed; prior-scale effects include sampling and terminal feature extraction.

2 Discussion

The experiments separate two main sources of error. Without chirality filtering, refinement followed by rescoring raised mean RMSD–PoseBusters success by 21.7–26.5 percentage points while mean RMSD-only selected-pose success was 1.0–3.7 points higher. Under the primary chirality-filtered selector, the RMSD–PoseBusters gain was 15.7–23.2 points. Matched-index analyses support geometric repair within the evaluated subset, whereas the aggregate change also includes reselection. After refinement, the RMSD-only oracle–top-1 gap was largest on PhiBench-derived and OpenBind. This identifies ranking loss within the tested candidate banks, without establishing that every unselected near-native candidate is physically valid.

Candidate density helps explain the cohort differences. Near-native poses were common in the Astex and PoseBusters banks, but only 6.3% of OpenBind candidates were near-native despite an 87.9% oracle rate. Sparse banks are more sensitive to ranking errors, and adding candidates can worsen top-1 when an incorrect pose is promoted. Flexible and highly fragmented ligands showed the same pattern more often, although the smallest strata were noisy.

The eligibility-aware decomposition confirms that chirality exclusion is rare, whereas ranking failure remains substantial when an eligible near-native candidate exists (Supplementary Fig. S18).

Training overlap varied widely across cohorts. Exact ligand matches were associated with higher success in PhiBench-derived and FoldBench but lower success in Astex, PoseBusters, and OpenBind, and sequence-identity bins showed no consistent trend. We did not observe a consistent monotonic relationship between the measured relatedness strata and success. Cohort composition, sparse strata and incomplete ancestral-checkpoint exposure records prevent these descriptive analyses from excluding training-set effects.

EFF-Dock assumes a prepared, fixed receptor, one ligand state and conformer, and a supplied pocket center. It does not model pocket discovery, receptor motion, water rearrangement, ligand-state enumeration, binding, or affinity. The receptor parser and guidance model support a restricted set of recognized monatomic-metal profiles, but this does not constitute general coordination-chemistry modeling. Unsupported cofactors or metal clusters, macrocycles near the fragmentation boundary, and covalent complexes remain outside the demonstrated scope.

Training pocket centers and fragment templates came from crystal complexes. Generation was tested with perturbed centers on Astex and PoseBusters while refinement and ranking retained the original center; complete-pipeline wrong-pocket robustness was not tested. Sensitivity to the starting ligand conformer was not systematically evaluated. Rigid-fragment motion cannot correct errors within a fragment. The greedy fragmentation procedure can also depend on atom order and uses soft, rather than exact, cut-bond closure.

The external cohorts are not prospective target-discovery benchmarks. Astex and PoseBusters were used repeatedly during development, the PhiBench-derived and FoldBench cohorts are task adaptations, and OpenBind is concentrated in one protease family, with two flagged noncovalent approximations of covalent systems. These cases do not constitute covalent-docking validation. Sequence and ligand relatedness to docking training data is high in parts of the evaluation set, including the highest sequence-identity bin for all OpenBind complexes and substantial same-sample ligand–sequence overlap in FoldBench. In addition, 341 of 558 FoldBench complexes (61.1%) share a PDB accession with a docking-training sample; accession sharing alone does not establish identity of the protein–ligand complex. The relatedness analysis does not reveal a monotonic association with success, but it also does not establish independence from training exposure or broad target-family generalization.

The experiments do not isolate the contribution of fragmentation, higher-order features, Newton–Euler readout, auxiliary losses or dynamic contacts. The confidence diagnostics quantify rank correlation, near-native discrimination, regret, and empirical reliability within the evaluated banks, but they do not establish calibration under new candidate-generation distributions. The reported standard deviations describe three inference repeats on fixed cohorts, including conformer variation only where it was part of the executed protocol. They do not represent variability across retrained models or uncertainty over target populations. Only 22 distinct complexes satisfied the joint sequence-identity, ligand-similarity and exact-match restrictions, so the stringent analysis provides limited evidence about generalization.

The evaluated pipeline uses fixed trained weights with revised coordinate-to-feature handling, as documented in Supplementary Information, Sec. S1.2. Coordinate-only rescoring uses a scene-defined frame convention for linear fragments. Finite checks found small rigid-rotation and backend resid-

uals under this convention; manually changing unobservable fragment twist still changes the frozen network’s predictions. The finite network rotation tests do not prove exact equivariance of the complete preparation, sampling, refinement, and selection pipeline. These tests hold the prepared local template fixed. They do not establish invariance to reorienting a supplied conformer before preprocessing, which changes its fragment body-coordinate convention; training-time local-coordinate rotation augmentation does not provide an exact architectural guarantee for that separate transformation.

The interaction energy is a geometric refinement model, not a free-energy function. The confidence score is used for ranking within a pose bank and is not calibrated across datasets or candidate-generation distributions. RMSD and PoseBusters capture pose accuracy and common coordinate errors, but they do not establish binding or identify every valid binding mode.

EFF-Dock models ligand flexibility with rigid fragments and learns their pocket-conditioned motion using SE(3)-equivariant flow matching. Refinement and rescoring improved selected-pose physical validity, while the RMSD-only oracle gap exposed substantial remaining ranking loss within the tested candidate banks. Across the tested supplied-pocket benchmarks, the method produced high candidate coverage with a 100-pose budget. Future work should improve pose selection and extend the model to receptor flexibility and broader input preparation.

3 Online Methods

Bold symbols denote vector-, matrix-, or structured-state-valued quantities, with each role identified at first use; unbold italic symbols denote scalars unless stated otherwise. The symbols $\|\cdot\|_2$, $(\cdot)^\top$, $(\cdot)^{-1}$, $\times$, $\odot$, and $(\cdot)^+$ denote the Euclidean norm, transpose, type-appropriate inverse or reciprocal, three-dimensional cross product, channel-wise multiplication, and the mode-truncated spectral pseudoinverse defined for each inertia operator, respectively. For a finite set $\mathcal{S}$, $|\mathcal{S}|$ denotes its cardinality, $\mathbf{0}$ is the dimension-compatible zero vector, and $\mathbf{I}_3$ denotes the 3×3 identity matrix.

3.1 Coordinate convention and evaluated weights

The coordinate reconstruction in Eq. (2) uses an active local-to-world rotation, $\mathbf{x} = \mathbf{R}\mathbf{y} + \mathbf{T}$. Equivariance is defined conditional on a fixed prepared fragment-local template and its body-coordinate convention. Under a joint rigid transformation of the receptor and current ligand pose, the prepared local coordinates remain fixed while world coordinates transform as $\mathbf{x}' = \mathbf{Q}\mathbf{x} + \mathbf{a}$ and fragment rotations as $\mathbf{R}' = \mathbf{Q}\mathbf{R}$. Supplied conformers are centered fragment-wise without canonicalizing their local axes. Accordingly, a learned fragment-local vector $\mathbf{w}$ is injected into world-frame features as

$$\mathbf{h} = \mathbf{R}\mathbf{w}, \quad \mathbf{h}' = (\mathbf{Q}\mathbf{R})\mathbf{w} = \mathbf{Q}\mathbf{h} \quad (1)$$

under a joint world rotation $\mathbf{Q} \in \text{SO}(3)$. Source-local edge coordinates of the form $\mathbf{R}^\top(\mathbf{x}_j - \mathbf{x}_i)$ perform the inverse world-to-local map.

All measurements use fixed docking and confidence weights; no retraining, probability recalibration, or result-dependent thresholding was performed. Coordinate-to-feature conventions, their differences from training, and finite numerical verification are documented in the [Supplementary Information, Sec. S1.2](#).

3.2 Task definition and fragment pose representation

EFF-Dock takes a prepared fixed receptor $\mathcal{P}$, a heavy-atom ligand graph $\mathcal{G}_L$ in one specified chemical state, one supplied or generated ligand conformer $\mathcal{C}_L$, and an explicit pocket center $\mathbf{b} \in \mathbb{R}^3$. We denote their complete conditioning context by $\mathcal{C}_{\text{cond}} = (\mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \mathbf{b})$. It returns a finite set of ligand poses and a separate score used only to rank poses within that set. Target ligand coordinates, target-derived atom or fragment frames, RMSD, validity outcomes, and benchmark identity are excluded from deployment inputs. Reference-derived pocket centers are used only in experiments explicitly labelled reference-pocket redocking and are not interpreted as pocket discovery.

The ligand contains N heavy atoms partitioned into K rigid fragments, with $i \in \{1, \dots, N\}$ and $k \in \{1, \dots, K\}$ indexing atoms and fragments, respectively. The map $g(i)$ assigns atom i to a fragment,

and $\mathbf{y}_i \in \mathbb{R}^3$ is its local coordinate relative to that fragment’s geometric centroid. The generative state and atom reconstruction are

$$\mathcal{M}_L = \prod_{k=1}^K (\mathbb{R}^3 \times \text{SO}(3)), \quad \mathbf{z} = \{(\mathbf{T}_k, \mathbf{R}_k)\}_{k=1}^K, \quad \mathbf{x}_i(\mathbf{z}) = \mathbf{R}_{g(i)}\mathbf{y}_i + \mathbf{T}_{g(i)}. \quad (2)$$

Here $\mathcal{M}_L$ is the ligand pose space, $\mathbb{R}^3$ is the translation space, and $\text{SO}(3)$ is the group of proper three-dimensional rotations. For fragment k , $\mathbf{T}_k \in \mathbb{R}^3$ is its geometric-centroid state origin and $\mathbf{R}_k \in \text{SO}(3)$ is its rotation; $\mathbf{z}$ collects all fragment states, and $\mathbf{x}_i(\mathbf{z}) \in \mathbb{R}^3$ is the reconstructed coordinate of ligand atom i in the pocket-centered frame. This representation preserves intrafragment geometry while allowing fragments to translate and rotate relative to one another.

Ligands are standardized with RDKit [23]: stereochemistry is assigned before hydrogen removal, the largest connected component is retained, and formal charge and the supplied protonation state are preserved. For simplified molecular-input line-entry system (SMILES) input, the inference path constructs one seeded Experimental-Torsion Knowledge Distance Geometry version 3 (ETKDGv3) conformer [24], followed by at most 200 Merck Molecular Force Field (MMFF) iterations [25]; supplied structure-data file (SDF) or Tripos MOL2 coordinates are retained. The receptor is hydrogen-stripped and fixed, and all receptor and ligand coordinates are translated by $-\mathbf{b}$ so that the supplied pocket center is the origin. For geometry training, each deposited crystal ligand pose is fragmented in its bound conformation, and the resulting crystal coordinates define the fixed fragment-local coordinates used as model input and reconstruction templates. Prospective inputs instead use supplied or independently generated conformers; rigid-fragment motion cannot correct internal differences within a fragment. Exact input normalization and pocket provenance are given in the [Supplementary Information, Sec. S1](#).

3.3 Fragmentation and pocket-conditioned graph

Fragmentation cuts eligible heavy-atom single bonds outside rings while excluding planar carbonyl C–N/O and sulfonyl S–N linkages. Connected components define provisional rigid fragments, singleton components are merged into adjacent fragments, and the retained cut topology supplies fragment adjacency and local cross-fragment distance constraints. This indexed greedy rule is deterministic for a fixed atom ordering; exact eligibility, tie-breaking, and failure handling are specified in the [Supplementary Information, Sec. S2.1](#).

The receptor is cropped residue-wise around the supplied center. Geometry training samples a radius uniformly from 6 to 12 Å and applies isotropic center jitter with standard deviation 2 Å; held-out validation uses an 8 Å crop, and the external reference-pocket protocol uses a fixed 10 Å crop. The graph contains ligand atoms, ligand fragments, protein atoms, and residue virtual nodes. Typed static relations encode covalent structure, fragment membership and topology, protein hierarchy, and residue–fragment context, while protein-atom–ligand-atom contacts within 5 Å are rebuilt at every forward pass. Node chemistry, all ten directed relation types, crop fallbacks, and edge descriptors are listed in the [Supplementary Information, Sec. S3.1](#).

3.4 Equivariant fragment flow and Newton–Euler readout

After pocket centering, fragment translations are drawn from an isotropic Gaussian and multi-atom fragment rotations from the uniform distribution on $\text{SO}(3)$ [26]. Let $t \in [0, 1]$ denote dimensionless flow time, with subscripts 0 and 1 denoting the sampled prior and bound target states, respectively, and let $\sigma > 0$ be the translation-prior standard deviation in Angstrom. For a paired prior and target state, translations follow a linear path and rotations follow the shortest geodesic under a left-action convention:

$$\begin{aligned} \mathbf{T}_{k,0} &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3), & \mathbf{R}_{k,0} &\sim \text{Uniform}(\text{SO}(3)), \\ \mathbf{T}_{k,t} &= (1-t)\mathbf{T}_{k,0} + t\mathbf{T}_{k,1}, & \mathbf{R}_{k,t} &= \text{Exp}(t\boldsymbol{\omega}_k^*)\mathbf{R}_{k,0}, \\ \mathbf{v}_k^* &= \mathbf{T}_{k,1} - \mathbf{T}_{k,0}, & \boldsymbol{\omega}_k^* &= \text{Log}(\mathbf{R}_{k,1}\mathbf{R}_{k,0}^{-1}). \end{aligned} \quad (3)$$

Here $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3)$ is the three-dimensional Gaussian prior and $\text{Uniform}(\text{SO}(3))$ is the normalized Haar distribution over rotations. The maps $\text{Exp} : \mathbb{R}^3 \rightarrow \text{SO}(3)$ and $\text{Log} : \text{SO}(3) \rightarrow \mathbb{R}^3$ are the rotation exponential and principal logarithm under the axis-angle identification. The starred vectors $\mathbf{v}_k^*$ and

ω_k^* are the target translational and angular tangents, measured in Angstrom and radians per unit flow time, respectively. Singleton rotations are fixed to identity and contribute no supervised angular mode. At time t , $\mathbf{z}_t = \{(\mathbf{T}_{k,t}, \mathbf{R}_{k,t})\}_{k=1}^K$ denotes the interpolated ligand state. Conditional flow matching trains the vector field $u_\theta(\mathbf{z}_t, t, \mathcal{C}_{\text{cond}}, \sigma)$, with learned parameters θ , to return the fragment-wise predictions $\{(\hat{\mathbf{v}}_k, \hat{\omega}_k)\}_{k=1}^K$ and match the starred targets without integrating the generative ODE during training [15]. The prior-scale mixture, time distribution, rotation augmentation, and numerical path conventions are recorded in the [Supplementary Information, Sec. S2](#).

Equivariant trunk and Newton–Euler readout. All node types share six equivariant interaction layers with 384 scalar channels, 32 channels of each vector parity and 16 channels of each rank-two parity. Time and prior scale enter through a 128-dimensional conditioning embedding. Each layer combines normalized sender features, radial edge features and real spherical harmonics through order two using tensor products, gated residual aggregation and adaptive normalization. Static relations and rebuilt protein–ligand contacts use relation-specific edge features. Complete equations and dimensions are given in the [Supplementary Information, Sec. S3](#).

The output head predicts one polar atom-vector proposal $\mathbf{f}_i$ for each ligand atom. Its unit-weight Newton–Euler aggregation is a least-squares rigid-body velocity fit restricted to the retained rotational modes; these vectors are not physical forces or energy gradients. For fragment k , with atom set $\mathcal{A}_k$, center $\mathbf{T}_k$, and lever arm $\mathbf{r}_i = \mathbf{x}_i - \mathbf{T}_k$, the fragment field is obtained by

$$\begin{aligned}\hat{\mathbf{v}}_k &= \frac{1}{|\mathcal{A}_k|} \sum_{i \in \mathcal{A}_k} \mathbf{f}_i, & \boldsymbol{\mu}_k &= \sum_{i \in \mathcal{A}_k} \mathbf{r}_i \times \mathbf{f}_i, \\ \mathbf{J}_k &= \sum_{i \in \mathcal{A}_k} (\|\mathbf{r}_i\|_2^2 \mathbf{I}_3 - \mathbf{r}_i \mathbf{r}_i^\top), & \hat{\omega}_k &= \mathbf{J}_k^+ \boldsymbol{\mu}_k.\end{aligned}\tag{4}$$

Here $\hat{\mathbf{v}}_k$ is the predicted translational tangent, $\boldsymbol{\mu}_k$ is the moment of the atom vectors, $\mathbf{J}_k$ is the unit-mass geometric inertia about the fragment centroid, and $\hat{\omega}_k$ is the predicted angular tangent. The truncated pseudoinverse $\mathbf{J}_k^+$ removes geometric-inertia eigenmodes at or below 1% of the largest eigenvalue, and the same observable subspace is applied to the angular training target. This prevents singleton and nearly collinear fragments from receiving arbitrary angular velocities; the exact eigenspace construction is given in the [Supplementary Information, Sec. S3.2](#).

3.5 Objective and sampling

The primary losses are mean-squared errors (MSEs) on fragment translation and observable angular velocity. The geometry objective combines fragment translation and observable angular-velocity errors with two auxiliary terms:

$$\mathcal{L}_{\text{geom}} = \mathcal{L}_v + \lambda_\omega \mathcal{L}_\omega + \lambda_{\text{atom}} \mathcal{L}_{\text{atom}} + \lambda_{\text{DG}} \mathcal{L}_{\text{DG}}.\tag{5}$$

Here $\mathcal{L}_v$ is fragment translation MSE, $\mathcal{L}_\omega$ is angular MSE after projecting the target onto retained rotational modes, and $\mathcal{L}_{\text{atom}}$ compares reconstructed atom velocities. The distance-geometry term $\mathcal{L}_{\text{DG}}$ compares interfragment atom-pair distances after a one-step endpoint extrapolation, with a t^2 -weighted valid-complex reduction. Unobservable terms contribute zero. The final continuation uses weights $(\lambda_\omega, \lambda_{\text{atom}}, \lambda_{\text{DG}}) = (8, 0.3, 3)$; definitions, normalizations and distributed reductions are given in the [Supplementary Information, Sec. S3.3](#).

The final continuation samples flow time from an 80% base distribution, a 10% uniform component on $[0, 0.3]$, and a 10% point mass at $t = 0$. The base distribution mixes uniform sampling with a late-biased logit-normal distribution. Its parameters and the prior-scale mixture are specified in the [Supplementary Information, Sec. S2](#).

At inference, sampled fragment priors are integrated from $t = 0$ to 1 using 10 late-biased Euler intervals in the primary protocol. For interval width Δt , translations receive additive increments and rotations are advanced by left multiplication with $\text{Exp}(\Delta t \tilde{\omega}_k)$, where the tilde denotes the guided field when guidance is enabled. Atom coordinates and dynamic protein–ligand contacts are rebuilt before every field evaluation. Candidate generation, raw-output preservation, and failure semantics are specified in the [Supplementary Information, Sec. S2.2](#).

3.6 Confidence-based pose selection

Pose generation and top-1 selection are separated. For each candidate, the fixed docking network is evaluated at $t = 1$, using recovered fragment frames and the source translation-prior scale, to recompute terminal ligand representations. A second equivariant network combines these with a fresh heterogeneous-graph embedding and applies four additional interaction layers. Invariantized node features and explicit protein-contact summaries are pooled by ligand-contact attention and type-wise mean-maximum aggregation. Atom and pose heads are trained with displacement and RMSD regression, 2 Å success classification, pairwise ranking, and a success-focused listwise objective. For a candidate set $\{\mathbf{z}_S^{(m)}\}_{m=1}^M$, the deployed selector is

$$\mathcal{E} = \begin{cases} \{m : q_m = 1\}, & \text{if any } q_m = 1, \\ \{1, \dots, M\}, & \text{otherwise,} \end{cases} \quad m^* = \arg \min_{m \in \mathcal{E}} c_\phi(\mathbf{z}_S^{(m)}, \mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \sigma). \quad (6)$$

Here q_m denotes compatibility with the input-defined tetrahedral stereocenters. Ties follow saved candidate order. The unfiltered argmin is reported separately as a diagnostic condition. The confidence network’s nonnegative predicted-RMSD score is c_ϕ , with learned parameters ϕ ; m^* is the selected candidate index. The prior scale enters this score through the recomputed docking representations. The score is interpreted only as a within-candidate-set ranking signal and not as calibrated RMSD, uncertainty, binding probability, or affinity. The energy-restraint reference and confidence-input template are rebuilt using the recorded generation preparation and conformer seed. Refinement preserves each saved candidate’s fragment-internal coordinates. For rescoring, candidate coordinates are mapped to fragment frames by proper Kabsch fitting for templates of rank at least two and an input-indexed, scene-defined convention for lower-rank fragments. This resolves the recovery convention without making the frozen network invariant to unobservable fragment twist; finite numerical checks and their limits are reported in the [Supplementary Information, Sec. S1.3](#). The complete readout and confidence objective are given in the [Supplementary Information, Sec. S4](#).

3.7 Differentiable guidance and post-sampling refinement

For ligand coordinates $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$, we define a differentiable guidance energy

$$\begin{aligned} E_G &= E_{\text{phys}} + E_{\text{int}}, \\ E_{\text{phys}} &= E_{\text{bond}} + E_{\text{angle}} + E_{\text{proper}} + E_{\text{chiral}} + E_{\text{planar}} \\ &\quad + E_{\text{LJ}} + E_{\text{steric}} + E_{\text{obs}}, \\ E_{\text{int}} &= E_{\text{hyd}} + E_{\text{HB}} + E_{\text{chg}} + E_\pi + E_{\text{cat-}\pi} + E_{\text{XB}} + E_{\text{metal}}. \end{aligned} \quad (7)$$

The cut-interface terms restrain only bond, angle, proper-torsion, chiral, and planar geometry that spans independently moving fragments. The nonbonded terms include a soft-core Universal Force Field (UFF)-style Lennard-Jones profile on admitted interfragment ligand–ligand and protein–ligand pairs, a compact receptor-overlap barrier and repulsion-only obstacles for unresolved receptor chemistry [27]. The seven interaction terms describe hydrophobic contacts, directional hydrogen bonds, screened formal-charge groups, pi stacking, cation–pi contacts, halogen bonds, and typed metal coordination [28, 29, 30, 31, 32]. These bounded geometric preferences do not define binding free energy, affinity, or a molecular-dynamics Hamiltonian.

Compact radial and angular gates use a clipped quintic smoothstep with zero first and second derivatives at its endpoints. This cutoff envelope joins constant regions with continuous energy and first and second derivatives; it is not a physical force law. Energy forms, occupancy aggregation, typing conventions and numerical constants are specified in the [Supplementary Information, Sec. S5](#).

The coordinate gradient $\nabla_{\mathbf{x}_i}$ gives the guidance force $\mathbf{F}_i = -\nabla_{\mathbf{x}_i} E_G$, which is distinct from the learned atom-vector proposal $\mathbf{f}_i$. For atom mass m_i , the fragment mass is $m_k^{\text{frag}} = \sum_{i \in \mathcal{A}_k} m_i$, the center of mass is $\mathbf{c}_k = (m_k^{\text{frag}})^{-1} \sum_{i \in \mathcal{A}_k} m_i \mathbf{x}_i$, and the resultant is $\mathbf{F}_k = \sum_{i \in \mathcal{A}_k} \mathbf{F}_i$. The vector $\boldsymbol{\tau}_k = \sum_{i \in \mathcal{A}_k} (\mathbf{x}_i - \mathbf{c}_k) \times \mathbf{F}_i$ is the center-of-mass torque, and $\mathbf{I}_k$ is the corresponding mass-weighted inertia tensor. The translational and angular guidance directions are

$$\mathbf{d}_k^\omega = \mathbf{I}_k^+ \boldsymbol{\tau}_k, \quad \mathbf{d}_k^T = \frac{\mathbf{F}_k}{m_k^{\text{frag}}} + \mathbf{d}_k^\omega \times (\mathbf{T}_k - \mathbf{c}_k). \quad (8)$$

Here $\mathbf{d}_k^T$ and $\mathbf{d}_k^\omega$ are guidance directions, $\mathbf{I}_k^+$ is the mass-inertia truncated pseudoinverse, and $\mathbf{T}_k - \mathbf{c}_k$ corrects for the offset between the fragment’s geometric-centroid state origin and its center of mass. In read-only mode, the energy components and projected directions are recorded as a guidance trace and coordinates are unchanged. For guidance experiments, let $(\Delta\mathbf{v}_k, \Delta\boldsymbol{\omega}_k)$ denote the mass-inertia directions after RMS matching to the learned atom velocity, late-time interval ramping, and the translation, rotation, and atom-displacement caps. The guided field is

$$\tilde{\mathbf{v}}_k = \hat{\mathbf{v}}_k + \Delta\mathbf{v}_k, \quad \tilde{\boldsymbol{\omega}}_k = \hat{\boldsymbol{\omega}}_k + \Delta\boldsymbol{\omega}_k, \quad (9)$$

where η controls the pre-cap guidance strength; the exact interval average, normalization, zero-direction rule, and caps are given in the [Supplementary Information, Sec. S5.9](#). This inference-time modification is conceptually related to guidance of flow-matching vector fields, although the normalization and fragment projection used here are method specific [33]. In the optional post-sampling refinement mode, a saved candidate is updated only on the fragment state. At refinement iteration r ,

$$\mathbf{T}_k^{(r+1)} = \mathbf{T}_k^{(r)} + \alpha_r \bar{\mathbf{d}}_k^T, \quad \mathbf{R}_k^{(r+1)} = \text{Exp}(\alpha_r \bar{\mathbf{d}}_k^\omega) \mathbf{R}_k^{(r)}, \quad (10)$$

Here $(\bar{\mathbf{d}}^T, \bar{\mathbf{d}}^\omega) = \text{Cap}(\mathbf{d}^T, \mathbf{d}^\omega)$ are capped increments and α_r is the accepted line-search step size. The cap first bounds each fragment translation and rotation separately, then jointly rescales both until the reconstructed atom displacement meets its bound. Writing $E_r = E_G(\mathbf{x}^{(r)})$, the proposed translation, rotation, and induced atom displacement are bounded and α_r is reduced by monotone backtracking until a finite non-increasing guidance energy is obtained. The refinement is bounded and stops on convergence, line-search failure, or a sustained energy plateau; the numerical safeguards and termination criteria are given in the [Supplementary Information, Sec. S5.10](#). Raw candidates are preserved, and refined coordinates are stored as a post-processing branch. The learned network parameters remain unchanged in all three modes.

The guidance subsystem uses a typed 18 Å receptor shell, which can expose coordinates outside the 10 Å neural crop. This shell is not an additional neural input. Full typing rules, energy equations, mass-inertia projection, refinement safeguards, and permitted information are provided in the [Supplementary Information, Sec. S5](#).

3.8 Data curation and split

Training data were derived from PLINDER 2024-06/v2 [34]. The ordered curation procedure removes invalid, covalent, and ion-like ligands; applies interaction, experimental-quality, ligand-resolution, contact, and size criteria; and selects deterministic representatives by complex identity, ligand identity, and pocket70 community. Of 1,357,906 indexed rows, 50,297 complex-ligand records remained after filtering and deduplication, and 50,213 were successfully preprocessed. The complete funnel, thresholds, missing-value policy, and 84 quarantined failures are reported in the [Supplementary Information, Sec. S6](#).

The docking continuation uses 47,277 training samples after input filtering, with 1,076 held-out identities. The confidence pipeline retains 43,092 training and 1,035 validation samples after its own input filters. The training sets share 43,067 sample IDs, with 4,210 docking-only and 25 confidence-only entries; neither is a subset of the other.

The historical external exclusion removed samples only when an exact ligand match and membership in an external pocket-community set occurred together. This procedure removed 904 samples; it does not enforce independent ligand or protein novelty, and broader sequence or ligand relatedness remains in the evaluation cohorts. Public membership lists record the included IDs, exclusion reasons, and set hashes; detailed accounting is given in the [Supplementary Information, Sec. S6](#).

3.9 Evaluation scope

Row-level splits can share repeated ligands or homologous proteins, so released-set relatedness is measured explicitly rather than inferred from split labels [34, 35, 36, 37]. For each external cohort, we report the receptor state, pocket source, ligand preparation, candidate budget, and selector. Astex and PoseBusters were used during development and are treated as descriptive benchmarks.

The geometry model uses the receptor crop, ligand graph, starting conformer, pocket center, and current sampled coordinates. The confidence model ranks the generated candidates from the same

task inputs. The interaction energy may additionally read a larger receptor shell, but it has no access to the crystal pose, RMSD, or validity labels. Raw coordinates are saved before refinement.

3.10 Model training and selection

The docking model was trained sequentially for 200,000 base updates, 100,000 geometry-fine-tuning updates and 50,000 early-time-replay updates. The first two stages used global batch size 256; the final stage used 64. Optimizer and scheduler states were reset between stages, while model and available EMA states were retained. The geometry-stage initializer was selected by median ligand RMSD on internal validation rollouts, and the evaluated weights are the prespecified terminal EMA of the final continuation. This continuation used AdamW with global batch size 64 across four GPUs, peak learning rate 2×10^{-5} , weight decay 0.01, gradient clipping at 1.0 and EMA decay 0.999 [38]. The learning rate warmed up for 1,000 updates, remained constant through update 40,000 and decayed by cosine scheduling to 2×10^{-6} over the final 10,000 updates. The confidence model was initialized through two preceding training stages. An initial model trained on 4,096 complexes was selected at update 42,500 by success-head-ranked accuracy on a 256-complex internal-validation subset. It initialized a 50,000-update symmetry-aware adaptation, whose terminal state initialized the subsequent raw-and-refined training run. The evaluated weights were selected at update 70,000 of that 100,000-update continuation, using predicted-RMSD ranking. Its released checkpoint was selected by confidence-ranked top-1 RMSD success on the fixed 1,035-complex internal validation bank. The final docking and confidence checkpoints were selected by the specified internal-validation rules, rather than their external benchmark results. Earlier development and checkpoint lineage are described in the [Supplementary Information, Sec. S6.4](#). Both released checkpoints are fixed throughout evaluation; terminal docking features are recomputed separately for confidence scoring. Confidence-model construction, pose-bank composition, and internal selection are detailed in the [Supplementary Information, Sec. S4](#) and [Supplementary Information, Sec. S6.4](#).

3.11 Benchmarks and comparison protocol

The evaluation uses five complementary cohorts. The Astex Diverse Set contains 85 curated complexes [39], and PoseBusters v2 contains 308 complexes with detailed coordinate-validity checks [10]. The PhiBench-derived cohort comprises 206 locally selected complexes from the public PhysDock archive; identity with the published 206-case cohort has not been established [40]. FoldBench-Pocket contains 558 released protein-ligand interfaces, including a 66-interface post-June 2024 slice [41]. OpenBind contains 925 EV-A71/CVA16 2A-protease complexes and is an auxiliary single-family cohort, including two flagged noncovalent approximations of covalent systems [42].

All five cohorts are evaluated by holo-receptor, reference-pocket redocking. The pocket center is computed from receptor residue nodes within 8 Å of the reference ligand, and the reference ligand is then removed from the model input.

The primary setting uses 100 candidates, 10 late-biased Euler steps, $\sigma = 2$ Å, a 10 Å receptor crop, and no inference-time guidance. Each cohort uses the same trained weights with inference seeds 42, 100042 and 200042. Astex and PoseBusters fix the ETKDG seed at zero for generation; PhiBench-derived, FoldBench and OpenBind use the per-complex sampling seed for conformer generation, so their repeat variability also includes starting-conformer variation. Candidates are scored before and after bounded refinement. The primary selector removes chirality-mismatched candidates and then chooses the lowest predicted symmetry-aware RMSD; if none passes the chirality screen, it uses the unfiltered ranking. Failed complexes remain in the cohort denominator. Supplementary Figs. S4 and S5 report runtime and generation-process memory for the initial candidate-acquisition campaign; these measurements exclude the separately executed downstream correction stages. Sensitivity to generation inputs is analyzed in Fig. 6; supplementary guidance controls are described below.

3.12 Generation-input sensitivity and guidance controls

Sensitivity experiments use all 85 Astex and 308 PoseBusters complexes with the same three inference seeds and chirality-filtered selector. At translation-prior scale 2 Å, generation crop radii 6, 8, 10, 12, 14 Å are crossed with Gaussian center jitter of 0, 1, 2 Å per Cartesian axis. Separately, prior scales 1, 2, 4 Å are crossed with the same jitter levels at a 10 Å crop. Shared cells are evaluated once, and the primary

baseline is reused from the primary evaluation. The jitter random stream is independent of the pose prior, and matching seeds couple prior draws within a candidate budget. Refinement and confidence scoring always use the original center and a 10 Å crop; the docking-feature re-evaluation used for confidence scoring retains each candidate’s source prior scale.

Four budget/guidance conditions, including the shared primary baseline, cross 100 poses/ten steps or 40 poses/25 steps with unguided sampling or normalized energy guidance of strength $\eta = 2$. Both budgets use 1,000 learned pose-steps, but guidance and candidate processing add work, so equal pose-step counts do not imply equal runtime. Forty- and 100-pose banks are generated separately, without a nested-prior claim. All sensitivity and guidance conditions are descriptive external evaluations and do not choose a sampler, guidance strength or selector.

3.13 Endpoints and statistical analysis

The primary pose endpoint is confidence-selected, symmetry-aware heavy-atom RMSD below 2 Å after complete atom mapping. Secondary endpoints include top- q and oracle- q coverage and complete-case RMSD summaries. Physical validity uses PoseBusters 0.6.5 in **redock** mode, with a reference-molecule representation adapter for the three temporal cohorts as detailed in the [Supplementary Information, Sec. S7](#); pass-all is the conjunction of all 27 non-RMSD checks on the same selected pose [10]. RMSD–PoseBusters success requires both RMSD < 2 Å and PoseBusters pass-all.

Binary endpoints use the full cohort denominator; continuous summaries use successfully evaluated poses and report their coverage. Error bars are sample standard deviations across the three repeats unless stated otherwise. Supplementary Figs. S9 and S14 use paired 95% bootstrap intervals. Further evaluation details are given in the [Supplementary Information, Sec. S7](#).

3.14 Use of automated tools

OpenAI Codex and Anthropic Claude were used to assist with code review, analysis-script development, manuscript drafting, consistency checks and language editing. The authors reviewed the generated text against the source code and primary literature and retain full responsibility for the manuscript. These tools are not authors.

Data and Code Availability

PLINDER, Astex, PoseBusters, PhiBench, FoldBench, and OpenBind structures remain governed by their original distribution terms; this manuscript does not redistribute them. Source code, released weights, benchmark summaries, and reproducibility instructions are available from the [public EFF-Dock repository](#). Training membership lists, exclusion reasons, numerical figure inputs and analysis code are provided alongside model and evaluation provenance. Coordinate-level evaluation materials that are not already included in the public result tables will be distributed only where permitted by the licenses and redistribution terms of the underlying benchmark resources.

Author Contributions

J.S.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and writing—original draft. J.L.: conceptualization, methodology, resources, supervision, project administration, and writing—review and editing. Both authors are responsible for the final scientific content.

Competing Interests

J.L.’s affiliation with Arontier Co. is disclosed in the author information.

Supplementary Information

The Supplementary Information specifies inputs and pockets (Section S1), fragmentation and sampling (S2), architecture and objectives (S3), confidence scoring (S4), energy and refinement (S5), data curation and training (S6), and evaluation (S7).

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